BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: MRD-IX-21(A)

Retyped faithfully from the original board paper — same questions, no changes.

Section "A"

Time Allowed: 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** MRD-IX-21(A)

Note: Note: There are three sections in this paper i.e. A, B & C. Instructions: Please attempt section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, overwriting or multiple circle shading.

Q1. Write the correct option i.e. A, B, C or D in the empty box provided opposite to each part. Cutting, erasing and overwriting will not be awarded any marks.

Q1. The additive inverse of √5 is .....

(A) $-\sqrt{5}$ (B) $1/\sqrt{5}$

(C) $\sqrt{-3}$ (D) -5

Q2. The matrix [4 0; 0 −12] is .....

(A) Scalar matrix (B) 2x3 matrix

(C) a diagonal matrix (D) None of these

Q3. Base of common log is .....

(A) e (B) π

(C) 5 (D) 10

Q4. L.C.M of $(a-b)^4$ and $(a-b)^3$ is.....

(A) (a-b) (B) $(a-b)^7$

(C) $(a-b)^4$ (D) $(a-b)^3$

Q5. Conjugate of $3 - \sqrt{5}$ is.....

(A) $-3 - \sqrt{5}$ (B) $-3 + \sqrt{5}$

(C) $3 + \sqrt{5}$ (D) $\sqrt{3} + \sqrt{5}$

Q6. Factor of $x^2+2x-24$ are.....

(A) x+4, x-6
(B) x-4, x+6
(C) x+3, x-8
(D) x+8, x-3

Q7. $\sqrt{(2x+1)} - 5 = 4$ is a .....

(A) linear equation
(B) cubic equation
(C) radical equation
(D) Quadratic equation

Q8. In ΔABC, if ∠A ≅ ∠B then bisector of .....divides, the ΔABC into two congruent triangles.

(A) ∠A
(B) ∠B
(C) ∠C
(D) Anyone of its angle

Q9. The point (-6,-6) is located in .....

(A) quadrant I
(B) Quadrant II
(C) quadrant III
(D) Quadrant IV

Q10. The angle bisectors of a triangle meet at a point which is equidistant from .....of the triangle.

(A) the vertices
(B) the sides
(C) mid points of sides
(D) all of these

Q11. In ΔKLM, m∠K=45°, m∠L=55° and m∠M = 80°, which one of the following is the longest side?

(A) KL
(B) LM
(C) KM
(D) None of these

Q12. In a mathematics class with 32 students, the ratio of girls to boys is 5 to 3. How many more girls are there than boys?

(A) 2
(B) 12
(C) 8
(D) 20

Q13. The centroid of a triangle divides the medians into the ratio of.....

(A) 2:1
(B) 3:1
(C) 4:1
(D) 5:1

Q14. Diagonal of a rectangle measures 6.5cm. If its width is 2.5cm, its length is .....

(A) 4cm
(B) 9cm
(C) 6cm
(D) 3cm

Section "B"

Time Allowed: 2 Hours 40 Minutes	**Maximum Marks:** 36

Q2. Attempt any NINE of the following parts. All parts carry equal marks.

i. If A = [2 0; −3 2], B = [1 −1; −1 3] find (i) AB (ii) BA

ii. Simplify $3\sqrt{(54x^3y^3z^2)}$

iii. Simplify with the help of logarithm: $(78.41)^3\sqrt{142.3} / (4\sqrt{0.1562})$

iv. Find the value of $x^3 - 1/x^3$ when $x - 1/x = 3/2$

v. Factorize the following (i) x^8+x^4+1 (ii) x^2+x-12

vi. Find H.C.F by division method $2x^5-4x^4-6x$ and $x^5+x^4-3x^3-3x^2$

vii. A man is 41 years old and his son is 9 years old. In how many years will the father be three times as old as the son.

viii. Draw the graph of the equation $y - 2x = 6$.

ix. Prove that $((4^5\cdot64^3\cdot2^3)/(8^3\cdot(128)^2))\text{^}(1/2) = 2$

x. Solve $\log_x 0.25 = 2$

xi. Simplify: $(x-y)/(x+y) - (x^2-2y^2)/(x^2-y^2)$

xii. Solve the following radical equation: $\sqrt{(2x-7)} + 8 = 11$

Section "C"

Maximum Marks: 24

> Note: Answer any THREE questions. All questions carry equal marks.

Q3. Prove that A(-4, -3), B(1, 1) and C(6, 11) are collinear.

Q4. Prove that any point on the bisector of an angle is equidistant from its arms.

Q5. Prove that if two angles of a triangle are congruent, then the sides opposite to those angles are congruent.

Q6. Construct ΔPQR, draw their altitudes and verify their concurrency. m $P\bar{Q}$ = 4.5cm, m $Q\bar{R}$ = 3.9cm and m∠R=45°.